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QUIZ #4
(Theory of Computation)

A1: a) $L = L(G)$
$G = (V, \Sigma, P, S)$

The proof is close to being correct, but not quite.
When we say L^*, L^0 is also included in that
However this rule gives us no way of generating ε.
Hence we also need to add $S \to \varepsilon$, along with $S \to SS$
→ The combined new production can be written as $\{S \to SS \mid \varepsilon\}$

The grammar can be written as:
$$G' = (V, \Sigma, P \cup \{S \to SS \mid \varepsilon\}, S)$$

Since L^* can be produced by a CFG, it is a CFL.
⇒ CFLs are <u>closed</u> under STAR operation (Kleene closure)

b) $\text{suffix}(L) = \{v \mid uv \in L, \text{ for some } u\}$
A CFG for this suffix(L) can be generated as follows:
i) For each variable X in CFG for L, add a new variable X' to generate suffixes of strings generated by X as the start variable.
ii) for each $X \to YZ$, add:
$$X' \to Y'Z \quad \text{and} \quad X' \to Z'$$
iii) also add
$$X' \to a \mid \varepsilon \; ; \; X \to a$$
iv) New start symbol is S'
$$S' \to X$$

As it can now be represented as a CFG, suffix(L) is a CFL.

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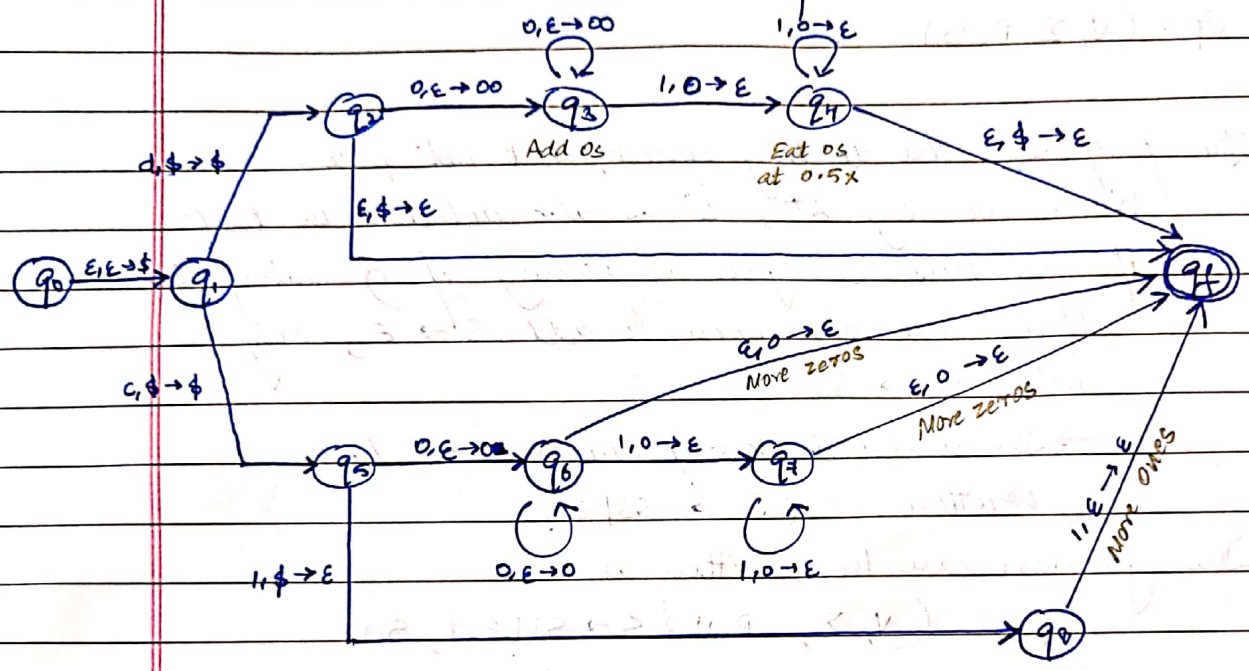
A2: $L = \{c0^m1^n \mid m \neq n,\ m,n \geq 0\} \cup \{d0^m1^{2m} \mid m \geq 0\}$

We can show that L is a DCFL iff we can construct a DPDA that accepts L.



★ Note that c and d act as a flag to direct us to either path and in their absence we would have to resort to guesswork, and hence it would not remain a DCFL.

⇒ Since a DPDA can be drawn, the language given in the question is indeed a DCFL.

A3: a) $S \to 0S1 \mid SS \mid \varepsilon$

$\Sigma = \{0, 1\}$

This language generates strings that are of the form known as "balanced parentheses". Consider 0 as '(' and 1 as ')' So the number of 0s and 1s are equal in the whole string but at no point can the number of 1s seen exceed the number of 0s. Also, string always starts with a 0 and ends with a 1, for obvious reasons.

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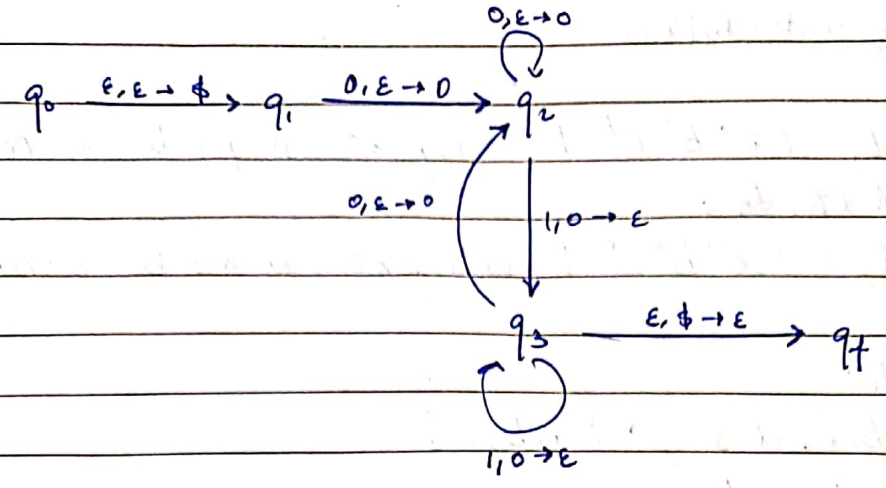
b)



Since a DPDA can be drawn for it, it is a DCFG.

A4: Let L be a CFL

WLOG,

L can be considered to be in CNF.

Hence, all the rules of L are of the following forms:

$$A \to BC$$
$$A \to a$$

where $A, B \in V$, $a \in \Sigma$

We can ignore rules of the form $A \to BC$ as they are not modified by homomorphism $h(L)$

Let us now construct a new grammar G'' for the language $L' = h(L)$ where $L' = L(G'')$.

G'' contains all prod^n rules from G of the form $A \to BC$.

For each rule of the form $A \to a$ in G, let $h(a)$ be $w \in \Sigma_h^*$

$w = w_1 w_2 \ldots w_n$ where $|w| = n$.

We add rules $A \to A_1 A_2$, $A_1 \to w_1$, $A_2 \to A_3 A_4$, $A_3 \to w_2, \ldots$

similar to what we do for converting CFG to CNF.

We have now created a CFG for L'.

$\Rightarrow$ Every CFL L is closed under homomorphism.

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b) Proof by contradiction:

Let $L_1 = \{a^m b^n c^p \mid m, n, p \geq 0, n = p\}$ be a DCFL

Let $L_2 = \{a^m b^n c^p \mid m, n, p \geq 0, m = n\}$ be a DCFL

Let h:

$\{a, b, c, d\} \rightarrow \{a, b, c\}$

s.t. $h(a) = a$

$h(b) = b$

$h(c) = c$

$h(d) = \varepsilon$

Let $L = (dL_1) \cup L_2$

$h(L)$

$\Rightarrow L_1 \cup L_2 = \{a^m b^n p^n \mid m, n \geq 0\} \cup \{a^m b^n p^n \mid m, n \geq 0\}$

this is CFL, but not deterministic

Guesswork reqd. at the beginning

$\Rightarrow$ DCFLs <u>not</u> closed under homomorphism.

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